



Cake Digital Bank: Creating a 'Next GenAI Bank' with Vertex AI

Get started for free

GOOGLE CLOUD RESULTS

- ✓ Enhances customer retention by reducing product development cycles from months to days
- ✓ Boosts mission to become a 'Next GenAI Bank' by creating 50 AI models in house with [Vertex AI](#), [Spanner](#) and [BigQuery](#).
- ✓ Supports regulatory compliance through AI biometric recognition system used by 15M+ customers
- ✓ Enables growth strategy through AI-driven digital ecosystem with partners from entertainment to insurance

In its mission to become Vietnam's leading digital bank, Cake Digital leveraged Vertex AI, in particular Gemini Flash, to create an omni-channel chatbot that brought personalized services to

customers using natural language. Now it's creating a digital financial ecosystem with external partners from retail, ewallet, entertainment, insurance, and more as it seeks to become a 'Next GenAI Bank.'



Since launching in 2021, [Cake Digital Bank](#) has embraced a mission to become a 'Next GenAI Bank,' integrating AI across all aspects of its operations, from back office to risk management and customer experience.

Through seamless AI experiences, it can approve loans or credit card applications in two minutes. It can also identify fraudulent behavior from app interactions, help customers secure accounts with biometrics, and send personalized messages based on previous behaviors.

Cake Digital Bank decided to build its architecture on [Google Kubernetes Engine](#) (GKE) thanks to the seamless and unlimited autoscaling of its microservices solution. For its AI needs it leveraged [Vertex AI](#), for everything from developing eKYC (electronic know-your-customer) systems, AI biometric recognition to building a sophisticated chatbot for a superior user experience.

By creating a digital-only bank powered by AI, Cake Digital Bank is confident it could serve the growing financial needs of Vietnam's digitally savvy and aspirational society. It chose Google Cloud thanks to the leadership position in AI solutions among major providers, as well as sharing its dream of empowerment through technology.

Building its architecture on Google Cloud, Cake Digital Bank acquired 4.6M customers in just under 4 years and was able to scale its business by deploying a range of Google Cloud services such as GKE, Cloud Spanner, [Cloud Run](#) and [BigQuery](#).

"Our mission is to propel the digital finance future of Vietnam to popularize and facilitate people's access to financial solutions in a smart and easy way," says Tu The Hien, CTO, Cake Digital Bank. "Google Kubernetes Engine and Vertex AI, which enables us to forge next-generation financial offerings, combine to help us achieve our goal of becoming a 'Next GenAI Bank.'

“Our mission is to propel the digital finance future of Vietnam to popularize and facilitate people's access to financial solutions in a smart and easy way. Google Kubernetes Engine and Vertex AI, which enables us to forge next-generation financial offerings, combine to help us achieve our goal of becoming a 'Next GenAI Bank.'”

Jay Tu The Hien
CTO, Cake Digital Bank



Leading Vietnam's digital-banking future with Google Cloud

Cake Digital Bank identifies two key benefits of using Google Cloud services. The first is fast, reliable, and unlimited autoscaling, which speeds up the product development process and shortens time-to-market. The speed of releasing new features on Cake Digital Bank is calculated in days rather than in months, as it would be in a traditional banking model.

Next, through use of Vertex AI, alongside BigQuery as its machine learning-enabled data warehouse, the bank's data science team enjoys world-class tools to build AI models in diverse operational segments. These include researching new products, assessing credit applications, and recommending tailored products for each customer. To date, Cake Digital Bank has developed 50 AI models in house using the most advanced Google Cloud AI technologies.

A considerable challenge in Cake Digital Bank's evolution was meeting Vietnam's strict regulatory biometric requirements for customer authentication. With effect from July 2024, State Bank of Vietnam mandates end-to-end biometric verification for financial transactions. These requirements led Cake Digital Bank to leverage Vertex AI to develop and roll out an eKYC and biometric verification system called FaceAuthen, the sole solution in the Vietnamese market that has been certified with iBeta Quality Assurance. Today, the bank's FaceAuthen is being used by 15M+ users in Vietnam.

“Gemini Flash is a powerful tool for creating intelligent and efficient chatbots. One of our key goals is understanding customers, and Gemini Flash helps our chatbot to understand even complex language patterns, tones, and emotions, allowing it to provide the most appropriate recommendations to the customer.”

Tu The Hien

CTO, Cake Digital Bank

Additionally, Cake Digital Bank sought to improve customer service performance and productivity through a chatbot that handles natural, intelligent, and personalized conversations. Cake Digital Bank employed Vertex AI, using [Gemini Flash](#) as its fast and nimble generative AI solution, to create the chatbot that was awarded Best AI Technology Implementation by the Asian Banker in June 2024.

One of the factors Cake Digital Bank valued in Vertex AI was that it came with large language models (LLMs) such as PaLM 2 and Gemini, both of which it found excellent at understanding and generating natural text.

Thanks to these LLMs, Cake Digital Bank enabled its Gato chatbot to understand context and continuously learn from user interactions to improve performance over time. As such, Gato is able to provide personalized experiences to each user based on interaction history and preferences.

"Gemini Flash is a powerful tool for creating intelligent and efficient chatbots," says Tu. "One of our key goals is understanding customers, and Gemini Flash helps our chatbot to understand even complex language patterns, tones, and emotions, allowing it to provide the most appropriate recommendations to the customer."



Building a dynamic financial ecosystem for customer personalization

In the future, Cake Digital Bank intends to create an AI-driven digital ecosystem with leading partners from many fields such as: telecommunications, insurance, retail, ecommerce, entertainment to fine-tune its understanding of customer behavior in diverse sectors. This will enable it to provide customers with the most suitable products and services for their financial needs.

"Cake Digital Bank's development of a comprehensive digital financial ecosystem will help customers access financial services at any touch point with the bank, including external parties," says Tu. "That will open a world of possibility for customers and partners. Creative uses of Vertex AI, BigQuery, and Gemini Flash are crucial to this mission."

"Cake Digital Bank's development of a comprehensive digital financial ecosystem will help customers access financial services at any touch point with the bank, including external parties. That will open a world of possibility for customers and partners. Creative uses of Vertex AI, BigQuery, and Gemini Flash are crucial to this mission."

Tu The Hien

CTO, Cake Digital Bank

Launched in 2021, **Cake Digital Bank** by VPBank currently serves more than 4.6 million customers, providing a full range of personal financial services, including transfer, payment, savings, investment, consumer loans, and credit cards. With a strategy to become a 'Next GenAI Bank,' Cake uses AI to bring the best experience to customers. In 2024, The Asian Banker recognized Cake Digital Bank as Best AI Bank, while Euromoney named it Outstanding Digital Bank 2024.

Industry: Financial Services

Location: Vietnam

Products: [Vertex AI](#), [BigQuery](#), [Cloud Run](#), [Gemini Flash](#), [Google Kubernetes Engine](#)

Why Google	Products and pricing	Solutions	Resources	Engage
Choosing Google Cloud	Google Cloud pricing	Infrastructure modernization	Google Cloud Affiliate Program	Contact sales
Trust and security	Google Workspace pricing	Databases	Google Cloud documentation	Find a Partner
Modern Infrastructure Cloud	See all products	Application modernization	Google Cloud quickstarts	Become a Partner
Multicloud		Smart analytics		Events
Global infrastructure		Artificial Intelligence	Google Cloud Marketplace	Podcasts
Customers and case studies		Security	Learn about cloud computing	Developer Center
Analyst reports		Productivity & work transformation	Support	Press Corner
Whitepapers		Industry solutions	Code samples	Google Cloud on YouTube
Blog		DevOps solutions	Cloud Architecture Center	Google Cloud Tech on YouTube
		Small business solutions	Training	Follow on X
		See all solutions	Certifications	Join User Research
			Google for Developers	We're hiring. Join Google Cloud!
			Google Cloud for Startups	Google Cloud Community
			System status	
			Release Notes	



English

